The Fundamental Theorem of Calculus

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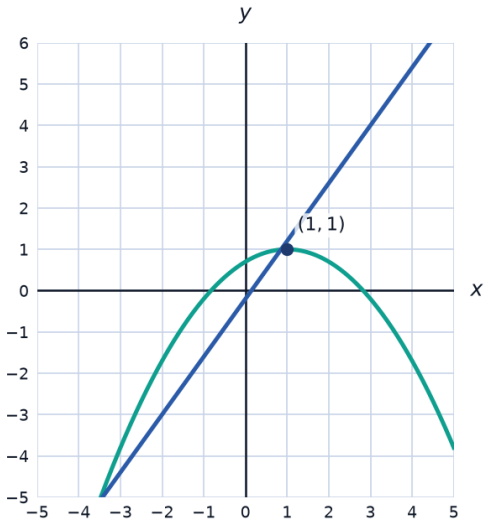
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: FTC Part 2: evaluating definite integrals

Evaluate $\int_0^3 (x^2 - 2x)\,dx$ using the Fundamental Theorem of Calculus.

2 [Foundational] • Target: FTC Part 1: accumulation functions

Let $g(x) = \int_0^x t^2\,dt$. Find $g'(x)$ using FTC Part 1.



3 [Foundational] • Target: FTC Part 1: accumulation functions

Let $g(x) = \int_0^{x^2} e^{-t}\,dt$. Find $g'(x)$ using FTC Part 1 combined with the Chain Rule.

4 [Foundational] • Target: Interpreting accumulation functions

A particle's velocity is $v(t) = 3 - 2t$ (m/s). Let $s(t) = \int_0^t v(\tau)\,d\tau$ be its displacement from $t = 0$. Find $s(4)$ and interpret its meaning.

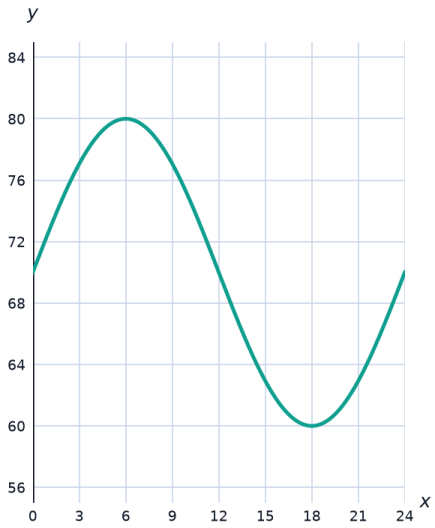
5 [Foundational] • Target: Interpreting accumulation functions

Given the graph of f consists of straight-line segments with $f(0) = 0$, and $g(x) = \int_0^x f(t)\,dt$, explain what feature of g's graph occurs where f crosses the x-axis, and where f has a local max.

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6 [Foundational] • Target: Average value of a function

The temperature in a city over a 24-hour period is modeled by $T(t) = 70 + 10\sin\left(\frac{\pi t}{12}\right)$ for $0 \le t \le 24$. Find the average temperature over this period.



7 [Foundational] • Target: Definite integrals with substitution-friendly integrands

Evaluate $\int_0^1 2x(x^2+1)^3\,dx$ using $u = x^2 + 1$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

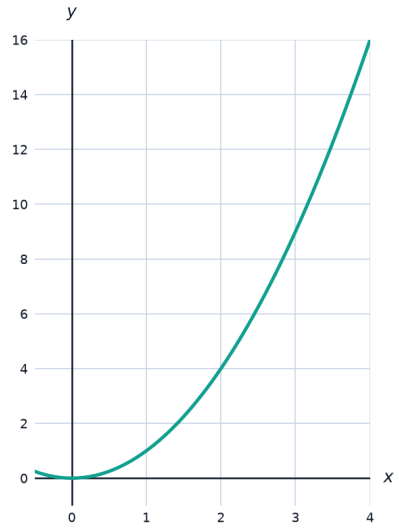
8 [Intermediate] • Target: The Mean Value Theorem for integrals

Find the value c guaranteed by the Mean Value Theorem for Integrals for $f(x) = x^2$ on $[0, 3]$, i.e. the point where $f(c)$ equals the average value found earlier.

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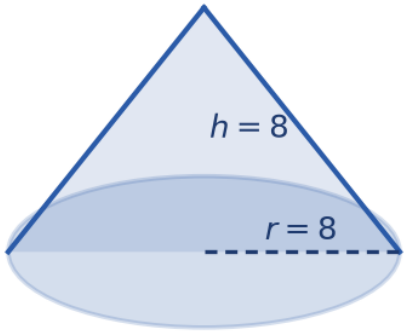
9 **[Intermediate]** • **Target: Visualizing FTC Part 1 as an accumulation function**

The graph shows $f(t) = t^2$ (the integrand from $g(x) = \int_0^x t^2\,dt$). Explain, using the graph, why g is always increasing, and why $g'(x) = f(x) = x^2$ matches the height of the curve at each x.



10 **[Intermediate]** • **Target: A capstone FTC application**

Water flows into a tank at rate $r(t) = 8 - t$ liters/minute for $0 \le t \le 8$ (the rate decreases as the tank fills). Using FTC, find the total volume of water added during the first 6 minutes, and explain why the rate being zero at $t = 8$ is consistent with the tank reaching capacity.



11 **[Intermediate]** • **Target: FTC Part 1 with more complex bounds**

Let $g(x) = \int_3^{2x} \ln(t+1)\,dt$. Find $g'(x)$ using FTC Part 1 and the Chain Rule.

12 **[Intermediate]** • **Target: A second average-value application**

A company's hourly production rate over an 8-hour shift is $p(t) = 50 + 10t - t^2$ units/hour, for $0 \le t \le 8$. Find the average production rate over the shift.

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13 **[Intermediate]** • **Target: Interpreting accumulation functions**

Level 2 · Medium Use Green's Theorem to evaluate $\oint_C (3y\,dx + 5x\,dy)$ where C is the counterclockwise circle $x^2 + y^2 = 4$.

14 **[Intermediate]** • **Target: Average value of a function**

[The Fundamental Theorem · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

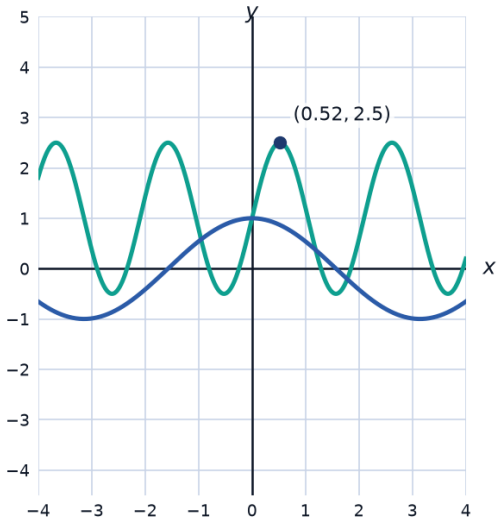
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Definite integrals with substitution-friendly integrands** • **Cross-Lesson: 7.01 Differential Equations and Slope Fields**

Solve the equation $z^3 + 8 = 0$, expressing all solutions in Cartesian form $a + bi$.

16 **[Challenging]** • **Target: The Mean Value Theorem for integrals** • **Cross-Lesson: 8.01 Area Between Curves**

Phone calls arrive at an emergency center according to a Poisson process with an average rate of 3 calls per minute. Find: (a) the probability that at least 2 calls arrive in a 1-minute period; (b) the probability that exactly 5 calls arrive in a 2-minute period.



17 **[Challenging]** • **Target: Visualizing FTC Part 1 as an accumulation function** • **Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods**

Level 3 · Advanced Let $f(x)$ be a function with $f(2) = 3$, $f'(2) = -1$, $f''(2) = 4$, and $f'''(2) = -12$. Write the third-degree Taylor polynomial $P_3(x)$ for f centered at $x = 2$, and use it to approximate $f(2.1)$.

18 **[Challenging]** • **Target: A capstone FTC application** • **Cross-Lesson: 1.01 Limits Graphically and Numerically**

Level 1 · Foundations Find $f_x(2, 1)$ and $f_y(2, 1)$ for the function $f(x, y) = \ln(x^2 + 3y^2)$.

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19 [Challenging] • Target: FTC Part 1 with more complex bounds • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

A continuous random variable X has probability density function $f(x) = 2e^{-2x}$ for $x \geq 0$ and 0 elsewhere. Find the cumulative distribution function $F(x)$ and determine the exact median of X.

20 [Challenging] • Target: A second average-value application • Cross-Lesson: 2.01 The Definition of the Derivative

Level 1 · Foundations Determine whether the vectors $\mathbf{u} = \langle 2, -3, 4 \rangle$ and $\mathbf{v} = \langle 6, 8, 3 \rangle$ are orthogonal, parallel, or neither.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: FTC Part 2: evaluating definite integrals

Let $g(x) = \int_0^x f(t)\,dt$, where f is continuous and negative for $t < -2$ and positive for $t > -2$. What feature does g have at $x = -2$?

a. An inflection point, since g is always increasing when its integrand is positive.

b. Neither -- accumulation functions never have local extrema.

c. A local minimum, since f changes sign from negative to positive at $t = -2$, so $g' = f$ changes from $-$ to $+$.

d. A local maximum, since f changes sign from negative to positive at $t = -2$, so $g' = f$ changes from $-$ to $+$.

22 [Foundational] • Target: FTC Part 1: accumulation functions

Let $g(x) = \int_0^x f(t)\,dt$, where f is continuous and positive for $t < 1$ and negative for $t > 1$. What feature does g have at $x = 1$?

a. Neither -- accumulation functions never have local extrema.

b. A local minimum, since f changes sign from positive to negative at $t = 1$, so $g' = f$ changes from $+$ to $-$.

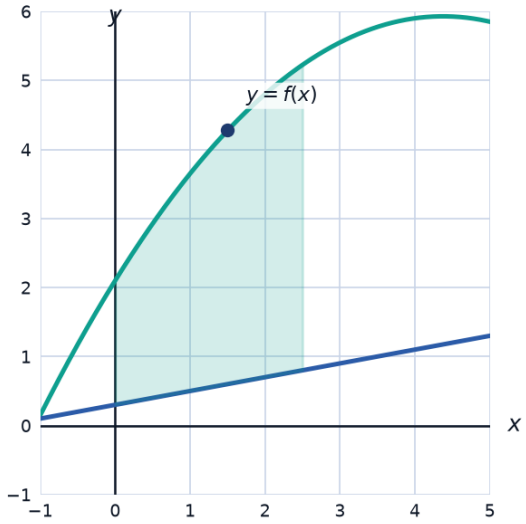
c. A local maximum, since f changes sign from positive to negative at $t = 1$, so $g' = f$ changes from $+$ to $-$.

d. An inflection point, since g is always increasing when its integrand is positive.

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23 [Foundational] • Target: Interpreting accumulation functions

For $F(x) = \int_a^x f(t)dt$ with f continuous, the proof of FTC Part 1 uses the Mean Value Theorem for Integrals to write $F(x+h) - F(x)$ as:



a. hf'(x)

b. $f(c) \cdot h$ for some $c \in [x, x+h]$

c. $f(x) \cdot (x+h)$

d. f(x+h)-f(x)

24 [Foundational] • Target: Average value of a function

If f has a jump discontinuity at $c \in (a, b)$, the accumulation function $F(x) = \int_a^x f(t)dt$ is still:

a. continuous at c (though not differentiable there)

b. discontinuous at c

c. differentiable at c

d. undefined at c

25 [Foundational] • Target: Definite integrals with substitution-friendly integrands

FTC Part 2: $\int_a^b f(x)dx = F(b) - F(a)$ where $F' = f$. Evaluate $\int_0^{\pi} \cos x \, dx$:

a. 0

b. 2

c. -2

d. 1

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26 [Foundational] • Target: The Mean Value Theorem for integrals

If f is continuous on $[a, b]$ and F is any antiderivative, FTC Part 2 requires no assumption that F is:

a. equal to f

b. unique — any antiderivative works, since they differ by a constant

c. bounded on $[a, b]$

d. differentiable everywhere

27 [Foundational] • Target: Visualizing FTC Part 1 as an accumulation function

FTC Part 1 shows every continuous function has an:

a. antiderivative

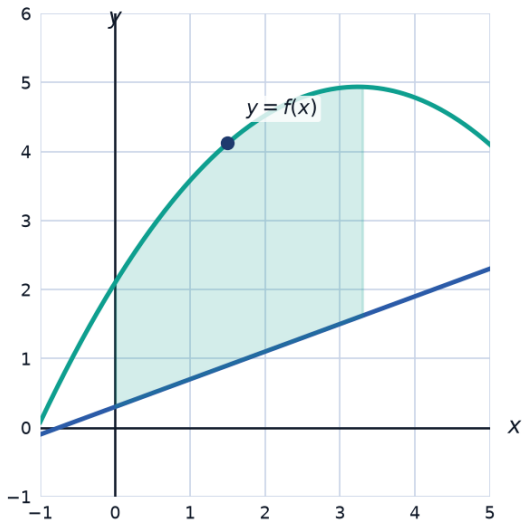
b. derivative everywhere

c. limit

d. inflection point

28 [Foundational] • Target: A capstone FTC application

Mean value theorem for integrals: $\int_a^b f\,dx = f(c)(b-a)$ for some $c \in (a, b)$. Requires f to be:



a. positive

b. integrable and bounded

c. continuous

d. analytic

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: FTC Part 1 with more complex bounds**

Mean Value Theorem for Integrals: if f is continuous on $[a, b]$, there exists $c \in (a, b)$ with $\int_a^b f(x)\,dx =$

a. $f(c)(b - a)$

b. $f'(c)(b - a)$

c. $f(b) - f(a)$

d. $f(c)$

30 **[Intermediate]** • **Target: A second average-value application**

For f continuous on $[a, b]$, the function $F(x) = \int_a^x f(t)dt$ is guaranteed to be:

a. differentiable on (a, b), with $F' = f$

b. only continuous, not differentiable

c. a polynomial

d. constant

31 **[Intermediate]** • **Target: FTC Part 2: evaluating definite integrals**

Proof of FTC: $F(x) = \int_a^x f$. Key step: $F(x + h) - F(x) = \int_x^{x+h} f(t)dt \approx f(x) \cdot h$ by:

a. MVT for integrals

b. L'Hôpital

c. product rule

d. chain rule

32 **[Intermediate]** • **Target: FTC Part 1: accumulation functions**

FTC Part 1 is proved by constructing:

a. any antiderivative

b. $G(x) = \int_a^x f\,dt$ and showing $G' = f$

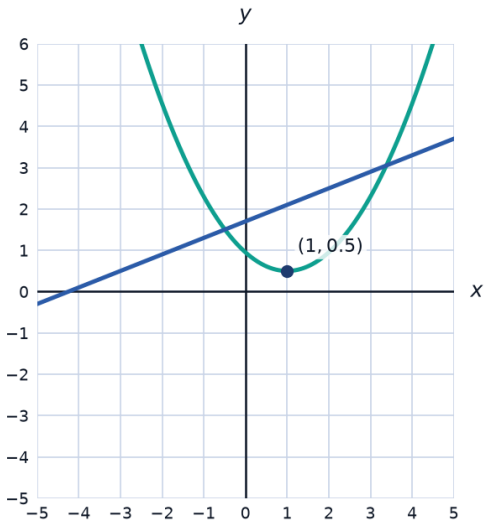
c. Taylor series

d. Riemann sums

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33 **[Intermediate]** • **Target: Interpreting accumulation functions**

FTC in higher dimensions: $\int_a^b ablaf \cdot d\mathbf{r} = f(b) - f(a)$. This is valid for:



a. any vector field

b. gradient fields (conservative) only

c. curl-free fields on simply-connected domains

d. divergence-free fields

34 **[Intermediate]** • **Target: Average value of a function**

Gradient theorem (FTC for line integrals): $\int_C ablaf \cdot d\mathbf{r} =$

a. $\iint f dA$

b. $f(\mathbf{r}(b)) - f(\mathbf{r}(a))$

c. $\oint f ds$

d. $abla^2 f$

35 **[Intermediate]** • **Target: Definite integrals with substitution-friendly integrands**

Let $F(x) = \int_2^x (t^2 - 1)dt$. According to the Fundamental Theorem of Calculus, what is the derivative of $F(x)$ with respect to x?

a. $x^2 - 1$

b. x^2

c. $x^2 - 1 - (2^2 - 1)$

d. $2x$

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36 **[Intermediate]** • **Target: The Mean Value Theorem for integrals**

IBP formula follows from FTC because $\int_a^b fg'\,dx = [fg]_a^b - \int_a^b f'g\,dx$ derives from:

a. Differentiating fg and applying FTC Part 2

b. L'Hôpital

c. Green's theorem

d. Cauchy-Schwarz

37 **[Intermediate]** • **Target: Visualizing FTC Part 1 as an accumulation function**

FTC allows computing $\int_0^1 x^{100}\,dx$ via:

a. L'Hôpital

b. power rule antiderivative

c. IBP

d. squeeze theorem

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: A capstone FTC application** • **Cross-Lesson: 1.01 Limits Graphically and Numerically**

Average value of f on $[a, b]$:

a. $\frac{f(a)+f(b)}{2}$

b. $\frac{1}{b-a}\int_a^b f\,dx$

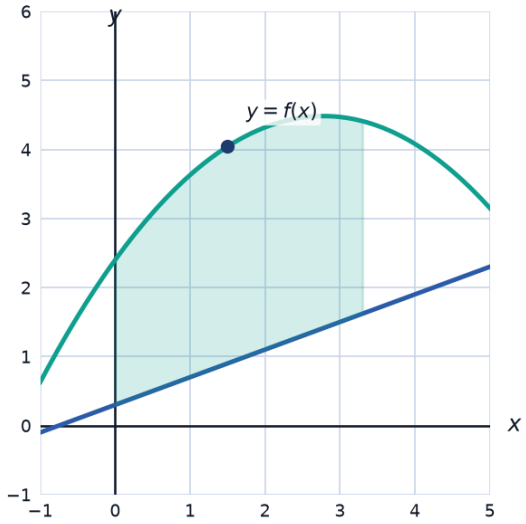
c. $\int_a^b f\,dx$

d. $f\left(\frac{a+b}{2}\right)$

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39 [Challenging] • Target: FTC Part 1 with more complex bounds • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

Fubini's theorem allows interchanging iterated integrals when:



a. both integrals converge

b. f is continuous

c. $\iint |f| < \infty$

d. $f \geq 0$

40 [Challenging] • Target: A second average-value application • Cross-Lesson: 2.01 The Definition of the Derivative

Net change theorem: $\int_1^4 v(t)dt$ gives the net displacement if $v(t)$ is velocity. For $v = 2t$:

a. 15

b. 8

c. 30

d. 4

41 [Challenging] • Target: FTC Part 2: evaluating definite integrals • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

Stokes' theorem $\iint_S (abla \times \mathbf{F}) \cdot d\mathbf{S} = \oint_C \mathbf{F} \cdot d\mathbf{r}$ generalises FTC to:

a. $\mathbb{R}^3$ surfaces

b. scalar fields

c. differential forms on manifolds

d. $\mathbb{R}^2$ only

42 [Challenging] • Target: FTC Part 1: accumulation functions • Cross-Lesson: 3.01 The Chain Rule

The Net Change Theorem $\int_a^b F'(x)dx = F(b) - F(a)$ requires F' to be:

a. continuous and positive

b. integrable on $[a, b]$

c. a polynomial

d. bounded and increasing

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43 [Challenging] • Target: Interpreting accumulation functions • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

A continuous function f and a constant a satisfy $\int_a^x f(t)\,dt = e^x - e^5$ for all x, with $a > 0$. Find a.

a. 25

b. -5

c. 10

d. 5

44 [Challenging] • Target: Average value of a function • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

MVT for integrals: $f(x) = x^2$ on $[0, 3]$. Value of c:

a. $\sqrt{3}$

b. $3/\sqrt{3}$

c. $3/2$

d. $\sqrt{3}/3$

45 [Challenging] • Target: Definite integrals with substitution-friendly integrands • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

MVT for integrals guarantees a value c called a:

a. critical point

b. root

c. mean point

d. zero

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